

In this journal entry, I will share my experiences during the ITAI 1371 midterm project. In this project, my group and I worked with the Telco Customer churn dataset. This is a split assignment between all 3 members of the group; I will explain my part in handling the Data cleaning and test split training of the dataset. The reason for this reflection is to help show my thinking process and what I learned as I cleaned my first dataset.

When I first started this project, I immediately saw an issue in the data. After I had set up all necessary libraries, I first printed out `df.info()`. As I was scrolling through the columns, the Total Charges column seemed incorrect, as it was labeled as an object instead of an integer, which could cause problems if we later used these numbers to train the AI. After seeing this, I decided to look into the Total Charges column first. I saw that there were 11 missing values for the column. This may not have been a big deal for a person looking at the database, but any AI would get thrown completely off by a string and stop working entirely, as it's considered a string in the eyes of the AI. I used pandas to transform the strings into numbers, and for the empty rows I used `errors='coerce'` to turn them into NAN instead of leaving them blank. I then decided to fill any NAN with 0, even though I was taught that the median is the safest option for guessing. I believe that these customers just haven't been billed, as I also saw low tenure in these customers as well, which most likely means they haven't paid yet.

Another issue I saw with the dataset in this project is the customer ID column. At first look, it may just look like it's just identifying customers; however, in a project, unnecessary numbers are exactly what machine learning builders look out for every day. Unnecessary numbers like these can completely throw a model off and are not useful to the AI.

Module 5 taught us about one-hot encoding, and as I was dropping this column, this somewhat reminded me of this idea of keeping data in categorical columns as 0's and 1's to make sure no category got an unfair advantage. If we don't, we may make the AI believe the column labeled 2 is more important than 1 when it's not, while customerID is not that type of problem, both don't let the AI see unnecessary numerical data.

Another part of the cleaning process I handled was creating the before and after histograms for Total Charges. I plotted both the raw and cleaned datasets side by side and noticed that, because only 11 out of 7000 rows are blank, the shape was very similar to the raw data. This is a similar realization I come to in all of my labs, where I start to understand that little things

matter just as much as the big picture. I have stated that small steps may not seem to have a major impact; however, these small changes do matter, as in the end it will increase the AI accuracy if you assume correctly, and without these small parts the entire machine could not learn properly. This shows that datasets basically look the same; they still have importance, no matter how small the change is, as it still affects the results.

After finishing cleaning the dataset, my next task was to split the data into 70/30 train/test sets. I remembered this process from the module 3 journal; this is also the module where I learned that every part of machine learning is important, and everything is just as important as the last. While thinking about the previous module, a question came to mind if someone did make a mistake in the code, would it affect other members of the group, and would they be able to notice? This is concerning, as if something can slip through multiple people and you decide that your data is ready for the AI and it's not, it could lead to hundreds of hours wasted by one mistake in the beginning.

This project showed me that even though a data engineer can have all the same problems as a different data engineer, the way they solve these problems can determine whether the AI is great or horrible. In my module 3, I asked the question of how to understand an AI and not just plug in numbers until it works. These last 2 modules and this midterm have helped me reach this goal by showing me the critical thinking necessary to make decisions on what data to keep and what to get rid of. This midterm and previous modules helped me understand how the first step of machine learning is done, and I hope to learn the next steps in the next modules.